

Board A — Earth-side control

Dual Adafruit Perma-Proto 1/2 carrier for the LilyGo T-Display-S3 and the EC11 menu encoder. Field I/O and power leave through edge screw terminals. This board is earth-referenced only.

Draft. Header pin order along each edge is taken from the carrier with the display up and USB-C on the left. The LilyGo pinmap photo is the module bottom, so its LEFT/RIGHT labels are mirrored here. Screw terminals are numbered **T1-T24** left to right on each edge (T1 nearest USB). Outer rails are as-built. Inner rails are proposed. Column span 7-18 is assumed. On-board jumper paths are documented elsewhere.

No floating copper. Do not land +5V_ISO, +12V_GD, -70V, +70V, or SW on Board A. MCU_GND is not -70V.

1. Placement

View with the display facing you. USB-C is on the left. The EC11 is on the right. Pin 12 of each header is nearest USB. Screw terminals are numbered **T1-T12** on the top edge and **T13-T24** on the bottom edge, left to right. The **top** edge starts at T1 (3V, GND, one NC, then PWM, DIR, limits, I²C). The **bottom** edge starts at T13 (5V, GND, five NC including T17-T19, then OPEN, LIM-L, ISENSE, HALL, 3V). The EC11 uses GPIO11-13 on the T-Display header directly; see EC11 on-board wiring.

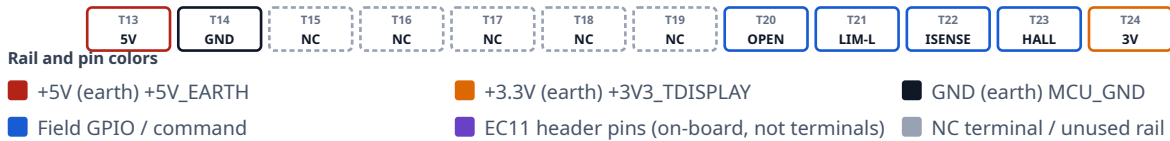
Board A — placement (first draft)

USB-C left · EC11 right · header pin 12 nearest USB · outer rails as-built · inner rails proposed

TOP edge · left → right



BOTTOM edge · left → right



Draft. Upper outer rails unused. Lower outer + unused; lower outer – is MCU_GND.
+5V_EARTH is the 5V screw terminal, not a proto + bus. Column span 7–18 is assumed.

Figure 1. Grid-true placement. Terminal IDs T1–T24 label each screw block left to right. Dashed gray bars are unused rails (as-built: both upper outer buses, and the lower outer +). Purple dots are T-Display header pins wired to the on-board EC11 (GPIO11–13). T17–T19 are NC; the encoder does not use those terminals.

2. Rails

Each half has four full-width buses. On the as-built unit both **upper outer** rails are unused, and the **lower outer +** rail is unused. The **lower outer –** rail is MCU_GND. Inner-rail nets below are still proposed. +5V_EARTH lands on the **5V** screw terminal and the T-Display **5V** pin; it is not on a proto + bus. Sensors / EC11 + should see +3V3_TDISPLAY from the module **3V** pin. Do not back-feed 3.3 V into the module.

Rail	Net	Diagram color
Upper outer –	unused (as-built; no jumper)	■ unused
Upper outer +	unused (as-built; no jumper)	■ unused
Upper inner +	+3V3_TDISPLAY (proposed)	■ +3.3V (earth)
Upper inner –	MCU_GND (proposed)	■ GND (earth)
Lower inner –	MCU_GND (proposed)	■ GND (earth)
Lower inner +	+3V3_TDISPLAY (proposed)	■ +3.3V (earth)
Lower outer +	unused (as-built; no jumper)	■ unused
Lower outer –	MCU_GND (as-built)	■ GND (earth)

3. What attaches

Power

Board A runs from the earth-side Mean Well **HDR-15-5 #1** supply (+5V_EARTH / MCU_GND). Land the PSU positive on **T13** (5V) and the PSU negative on **T14** (GND). Those terminals feed the T-Display **5V** pin and the board MCU_GND reference. USB-C on the module remains for bench programming. Do not back-feed 3.3 V into the module **3V** pin.

Screw terminals

Numbering follows Figure 1: left to right with the display facing you and USB-C on the left. T1 and T13 are nearest USB.

Top edge — T1-T12

Terminal	Label	GPIO / net	Attaches
T1	3V	+3V3_TDISPLAY	SS49E VCC, AS5600 VCC (from module 3V)
T2	GND	MCU_GND	Sensor, limit, and rocker returns
T3	GND	MCU_GND	Sensor, limit, and rocker returns. Lands R6 (PWM) and R7 (DIR) 10 kΩ pull-downs
T4	NC	—	Module NC. Unassigned. May be jumpered to MCU_GND.
T5	PWM	GPIO16	Board B 6N137 LED. On-board R6 10 kΩ to T3 MCU_GND
T6	DIR	GPIO21	Board B U4 LED. On-board R7 10 kΩ to T3 MCU_GND
T7	LIM-U	GPIO17	Upper NC limit to GND
T8	CLOSE	GPIO18	Close rocker (active-low, paralleled pair)
T9	SCL	GPIO44	AS5600 SCL
T10	SDA	GPIO43	AS5600 SDA
T11	GND	MCU_GND	Sensor, limit, and rocker returns; Board B LED return
T12	GND	MCU_GND	Sensor, limit, and rocker returns

Bottom edge — T13-T24

Terminal	Label	GPIO / net	Attaches
T13	5V	+5V_EARTH	Power in. Mean Well HDR-15-5 #1 (+) → T-Display 5V
T14	GND	MCU_GND	Power in. Mean Well HDR-15-5 #1 (-) / earth-side return
T15	NC	—	Module NC. Unassigned.
T16	NC	—	Module NC. Unassigned.
T17	NC	—	Unassigned. Same index as GPIO13; EC11 SW is on-board from the header (EC11 wiring).
T18	NC	—	Unassigned. Same index as GPIO12; EC11 DT is on-board from the header.
T19	NC	—	Unassigned. Same index as GPIO11; EC11 CLK is on-board from the header.
T20	OPEN	GPIO10	Open rocker (active-low, paralleled pair)
T21	LIM-L	GPIO3	Lower NC limit to GND
T22	ISENSE	GPIO2	ACS712-05B OUT after 10 kΩ/20 kΩ (≤ 3.3 V)
T23	HALL	GPIO1	SS49E OUT
T24	3V	+3V3_TDISPLAY	SS49E VCC, AS5600 VCC (from module 3V)

EC11 (on-board)

The menu encoder wires directly to T-Display-S3 header pins. It does not use screw terminals. Terminals **T17–T19** are NC on the as-built unit.

EC11 breakout	T-Display pin	GPIO / net
CLK	LEFT GPIO11	GPIO11
DT	LEFT GPIO12	GPIO12
SW	LEFT GPIO13	GPIO13
+	3V	+3V3_TDISPLAY
GND	GND	MCU_GND

Full notes, terminal clarification, and links: [EC11 on-board wiring](#).

On-board pull-downs

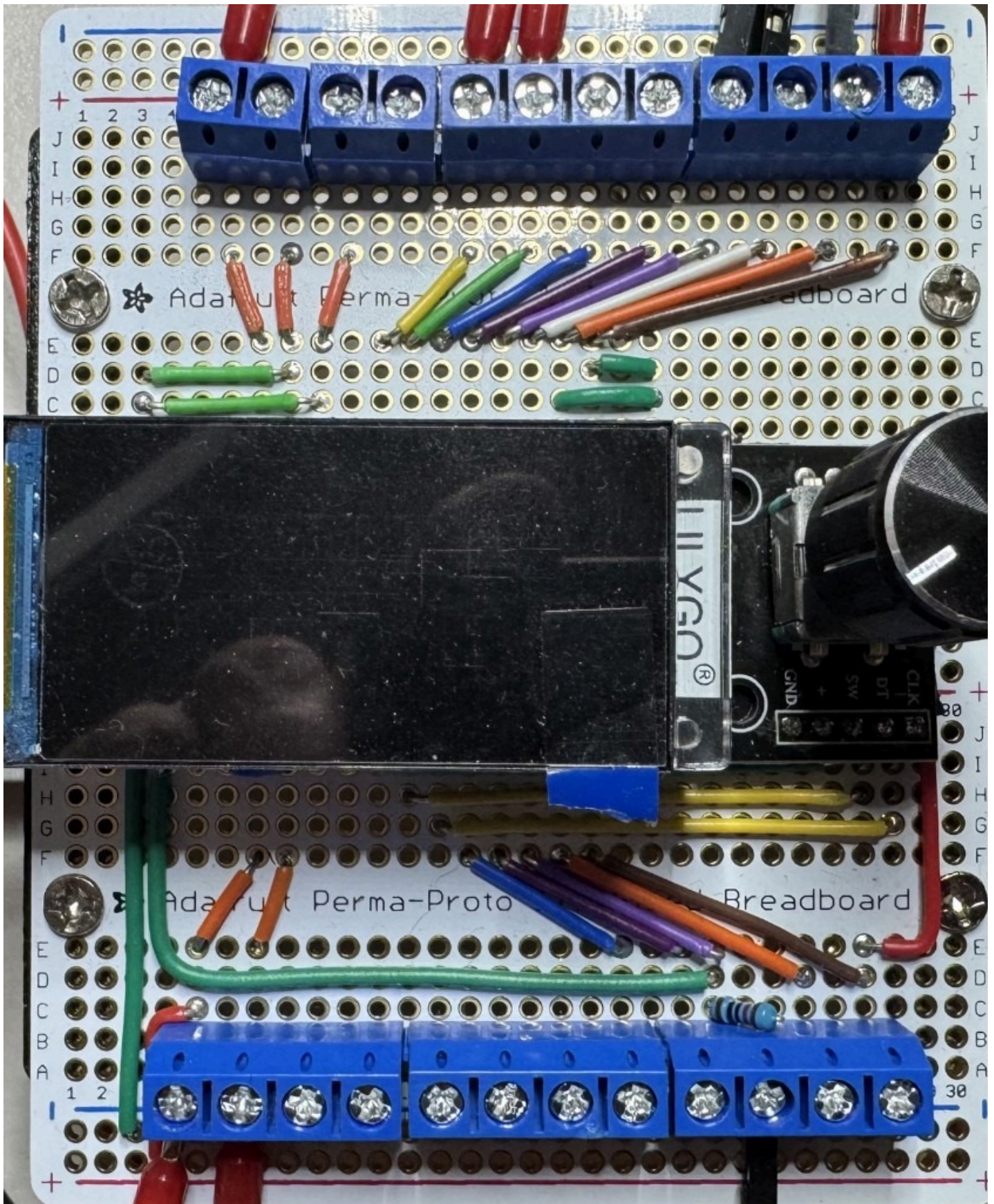
R6 (10 kΩ) sits between **T5** (PWM, GPIO16) and **T3** (MCU_GND). **R7** (10 kΩ) sits between **T6** (DIR, GPIO21) and the same T3 return. They hold the Board B optocoupler LEDs off if the T-Display is unpowered or resetting. Series LED resistors stay on Board B.

Board A → Board B

Keyed 3-pin earth-side cable from **T5** (PWM), **T6** (DIR), and a MCU_GND terminal (for example **T11** or **T14**). LED resistors stay on Board B.

4. As-built photo

Line art is the labeled figure. The photo shows the unit that was built.



Wired carrier. USB-C on the left, EC11 on the right.

Related

This page is a chapter of the Wiring Guide. Electrical wiring truth for the motor stack remains the gate-driver schematic, not the placement drawing.